

Deep Learning-Based Real-Time Mode Decomposition for Multimode Fibers

Yi An , Liangjin Huang , Jun Li, Jinyong Leng, Lijia Yang, and Pu Zhou 

Abstract—Mode decomposition (MD) is essential to reveal the intrinsic mode properties of multimode fibers (MMFs). Real-time MD provides a powerful tool to analyze the dynamics in MMFs. In this paper, we demonstrated that real-time MD can be achieved with the help of deep learning technique. We use large amounts of simulated beam intensity profiles of MMFs to train a convolutional neural network (CNN) and then evaluated this trained CNN on both simulation and experimental data. When testing on the simulated beam profiles, the averaged correlation between the reconstructed patterns and measured patterns is above 0.9842 and the decomposing rate can reach about 200 Hz. While for the experimental case, the averaged correlation is above 0.8896 and the decomposing rate for modal weights is 29.9 Hz, which is restricted by the maximum frame rate of the CCD camera. The results of both simulation and experiment show the superb real-time ability of the deep learning-based MD methods.

Index Terms—Fiber optics, machine learning, mode decomposition, optical fiber testing.

I. INTRODUCTION

MULTIMODE fibers (MMFs) have seen increasing applications in optical communication [1], imaging [2], sensing, nonlinear optics [3] and quantum optics [4]. Knowing the intrinsic properties of MMF will help push its development. There has been much related work such as spectrally and polarization resolved measurements [5]–[8] in the past few years. Besides, measuring the modal content of the fiber, which is known as mode decomposition (MD) technique [9], is also an vital and interesting topic. This technique provides the amplitude and phase information of each eigenmode in the MMFs so that the complete optical field and the beam properties associated with the field, e.g., wave front [10] and beam propagation factor [11] can be

inferred easily. Real-time MD will help characterize the mode dynamics of MMFs [12], [13] or monitor manufacturing mode-resolved components for mode division multiplexing [14].

Recent years, various MD approaches [9], [15]–[30] have been proposed, including the spatially and spectrally (S^2) method [15], correlation filter method (CFM) [16], [19] and numerical analysis method (NAM) [9], [21], [23], [30]. The S^2 method is based on both spatially and spectrally resolving the image of the output of the fiber under test, whose notable advantage is that no prior detailed knowledge of the fiber properties is required. But the measurement time of S^2 technique is much longer than other approaches, which is unsuitable for real-time modal analysis. CFM uses a set of transmission functions encoded in a computer-generated hologram (CGH) to complete MD process with a measuring rate up to 30 Hz [31], which has been also successfully applied for adaptive phase correction of dynamic beams of MMF [32]. However, the CFM needs more experiment efforts compared with NAM since the optical systems should be carefully aligned and calibrated. NAM might be the most economic and simplest way for modal analysis, which only needs a CCD camera to capture the beam intensity profile and then acquires the mode coefficient through numerical analysis. Among various NAMs, SPGD algorithm can achieve a monitoring MD rate of 9 Hz [23]. The decomposing rate is unsatisfying because the iterative searching process of this algorithm. In our previous work [27], [33], we have demonstrated that deep learning technique enabled by convolutional neural network (CNN) can be utilized to perform MD with an off-line process. That is, all of the investigated beam patterns of fibers are collected and stored previously, and then sent to the CNN for decomposition. Contrary to the off-line process, real-time decomposition, which means the beam pattern is acquired and directly fed into CNN without storage stage, is demanded in practice for time-varying patterns. In the Ref. [28], the authors also use CNN to help the MD for fiber beams and their scheme is different from ours. In the first step of this work [28], CNN is used to classify the investigated beam patterns to match a series of initial value matrices. Then the initial value is sent to the interior point (IP) algorithm for the MD result with an iterative process. The simulation results show that the iterative stage accounts above 90% of the total cost time, resulting in a real-time decomposing rate smaller than 0.5 Hz. In our scheme, CNN is utilized as a regression rather than classification tool, which means accurate MD result can be obtained directly from CNN with no additional iterative algorithm. Therefore, the decomposing rate can reach much higher.

Manuscript received September 30, 2019; revised January 17, 2020; accepted January 17, 2020. Date of publication February 14, 2020; date of current version April 13, 2020. This work was supported in part by the National Natural Science Foundation of China under Grant 61805280, in part by the Hunan Provincial Innovation Construct Project (2019RS3017), in part by the Natural Science Foundation of Hunan Province, China under Grant 2019JJ10005, in part by the Research Grants from College of Advanced Interdisciplinary Studies, National University of Defense Technology (JC18-07), and in part by the Open Research Fund of State Key Laboratory of Pulsed Power Laser Technology (SKL2018ZR06). (Yi An and Liangjin Huang contributed equally to this work.) (Corresponding Author: Pu Zhou).

The authors are with the College of Advanced Interdisciplinary Studies, National University of Defense Technology, Changsha 410073, China (e-mail: anyi0924@163.com; hlj203@nudt.edu.cn; lijun_gfkd@nudt.edu.cn; lengjiy@sina.com; ljyang@nudt.edu.cn; zhoup203@163.com).

Color versions of one or more of the figures in this article are available online at <https://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/JSTQE.2020.2969511

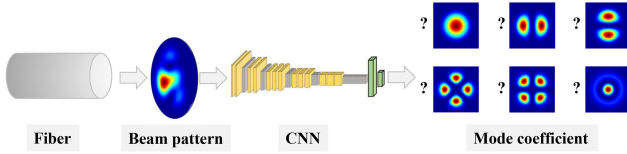


Fig. 1. The principle of deep learning-based MD scheme.

In this paper, we use only CNN to perform real-time MD in MMFs for the first time. Both the simulated and experimental beam intensity profiles are processed by a trained CNN in real-time and the results show a high accuracy and outstanding time efficiency. To acquire both the modal weights and modal phases, the decomposing rate in experiment can reach about 29.9 Hz for 3-mode case and 24 Hz for 6-mode case. This rate is close to the camera's maximum frame frequency, which is 30 Hz. When the trained CNN is only used to output the modal weights in experiment, the decomposing rate for two cases are both about 29.9 Hz, which is only restricted by the capturing rate of the CCD. To the best of our knowledge, the MD rate of our scheme is the fastest one among the current NAMs, which will be a vital and interesting supplementary for real-time MD.

II. METHOD

A. Principle of the Scheme

The principle of our scheme is illustrated in Fig. 1. The near-field intensity pattern of the beam emitting from the multimode fiber is fed into the trained CNN to directly estimate the mode coefficient without any iterative process. The beam pattern is captured by a CCD camera while the CNN is stored in a common computer, which are hidden in the figure. Since only a CCD and a computer are required to perform MD, the scheme is very simple and economic.

In this paper, two cases will be investigated respectively to verify the real-time ability of the trained CNN. In the first case, the fiber supports 3 eigenmodes, and we call it 3-mode case. The second case is the 6-mode case, and the involved fiber supports 6 eigenmodes.

To train the CNN, large amounts of samples, including near-field beam intensity images and their corresponding labels are needed. Here the training samples are acquired through simulation. The modal weights within the range of $[0, 1]$ and the modal phases within the range of $[-\pi, \pi]$ are randomly generated for computing near-field beam intensity images. The pixel values of these images are set in the range of $[0, 255]$ and the resolution of the image is 128×128 . As for the labels of the images, they are made up by the modal weights and modal phases based on two requirements. On the one hand, the value of labels should range from 0 to 1. On the other hand, one near-field beam pattern should match only one label to ensure the one-to-one mapping relationships. Modal weights satisfy these requirements very well so they become the first part of the label directly. However, modal phases demand some transforms. It is found that one pairs of mode coefficients whose sign of phases are totally opposite correspond to the same beam pattern, which is known as conjugated case [9]. Since the cosine value of opposite phases

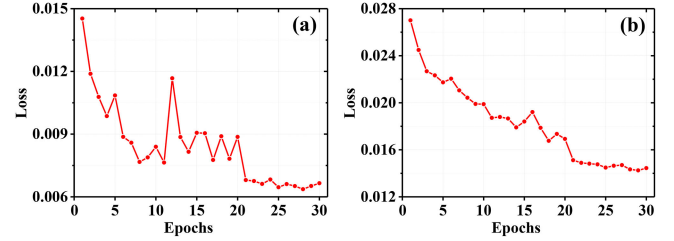


Fig. 2. The calculated loss as a function of epochs for two cases. (a) 3-mode case; (b) 6-mode case.

is the same, the modal phases are processed by cosine function and then linearly scaled to $[0, 1]$ as the second part of the label.

With the training samples, we modified VGG-16 model [34], which is a mature and widely used CNN architecture, to perform the training process. The first layer's filter size of VGG model is changed from $3 \times 3 \times 3$ to $3 \times 3 \times 1$ since the input is a single intensity pattern image while the last layer's Softmax function is replaced by Sigmoid for our regression problem.

We use a desktop computer with an Intel Core i7-8700 CPU and GTX 1080 GPU to train the CNN for two cases respectively. Due to the powerful parallelism calculation ability, GPUs are used to accelerate the computing speed. In the training procedure, input prepared sample image is passed through the layers of CNN and regressed into output vector which has the same length of label vector. The mean square error between the output and label is measured as the loss. Based on the loss and backpropagation algorithm, parameters of CNN are updated and utilized for the next training sample. The parameters of CNN are updated constantly from the first sample to the last sample, which means a training 'epoch'. In every training epoch, 100000 beam intensity images for 3-mode case or 200000 beam intensity images for 6-mode case are randomly generated to train the corresponding CNN respectively. The number of samples in the latter case is larger than the former case since the patterns become more complex. The learning rate is set to 0.01 in the first 20 epochs and 0.001 in the following epochs. The calculated loss of the CNN in the training procedure is shown in Fig. 2. We can find that after 30 training epochs, the loss of both two cases become steady, so we think these CNNs get convergence. The 3-mode case costs about 8 hours for training while 6-mode case costs about 16 hours. Then the trained CNN for 3-mode case or 6-mode case is utilized for real-time testing from simulation or experiment respectively. It is noted that the CNN only takes 30 epochs to gets convergence, which might be related with the large number of training samples in every training epoch and the set of learning rate. Another reason might be that the mapping between the beam pattern and their modal information is not complex so that the CNN can learn the relationship quickly.

The trained CNN can be utilized to perform non-iterative real-time modal analysis for near-field beam patterns either from simulation or acquired in the experiment. Taking a gray image of beam image as input, the modal weights can be predicted directly, which is just the first part of the output vector. As for the modal phases, they cannot be directly predicted, because only their scaled cosine value can be acquired from the output

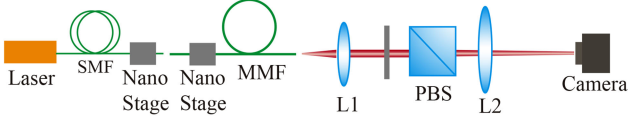


Fig. 3. Experimental setup of two cases. SMF, single mode fiber; MMF, multimode fiber; L, lens; HWP, half-wave plate; PBS, polarization beam splitter.

vector [27]. All the possible combinations based on the estimated cosine values are collected as candidates, and the final predicted phases can be determined by searching the maximum of the correlation between these candidates and the input intensity image, which can be defined as

$$C = \left| \frac{\int \int \Delta I_r(r, \varphi) \Delta I_m(r, \varphi) r dr d\varphi}{\sqrt{\int \int \Delta I_r^2(r, \varphi) r dr d\varphi \int \int \Delta I_m^2(r, \varphi) r dr d\varphi}} \right| \quad (1)$$

where $\Delta I_j(r, \varphi) = I_j(r, \varphi) - \bar{I}_j$ ($j = r, m$) and $\bar{I}_j$ is the corresponding mean value of measured beam intensity I_m or reconstructed I_r . The similarity between I_r and I_m can be judged by correlation C , whose maximum value is 1 when the reconstructed pattern is the same with the measured one.

B. Experimental Setup

Although two different cases are investigated in this paper, their experimental setup are similar, as illustrated in Fig. 3. The laser source is a narrow linewidth laser diode, followed by a single-mode fiber (SMF) to delivery light. The nano-stages are used to change the coupling conditions of the SMF and the multimode fiber (MMF). The beam emitting from MMF is captured by the Point Grey camera (CM3-U3-28S4M-CS) through a 4- f imaging system formed by lenses L1 and L2. The focal lengths for these two lenses are 8 mm and 400 mm respectively. The polarization beam splitter (PBS) between L1 and L2 is used to select one polarization component of the beam while the half-wave plate (HWP) is to change the polarization. Varying beam patterns can be obtained by rotating the HWP and these patterns are collected by the camera and sent to the trained CNN for real-time MD. The differences of two investigated cases lie on the laser source and the parameters of MMF. For 3-mode case, the MMF has 0.14 NA and 8.2 μm core diameter and the laser source works at 1073-nm wavelength. As for 6-mode case, the MMF has 0.2 NA and 7 μm core diameter and the laser source works at 910-nm wavelength.

III. RESULTS AND DISCUSSION

A. Simulated Results

Varying beam intensity patterns are generated continually and fed into the trained CNN for the simulation of real-time decomposition. To observe the MD result visually, the obtained mode coefficients are used to reconstruct patterns and it is found that whether in the 3-mode case or the 6-mode case the reconstructed patterns are both in high agreement with their corresponding input patterns.

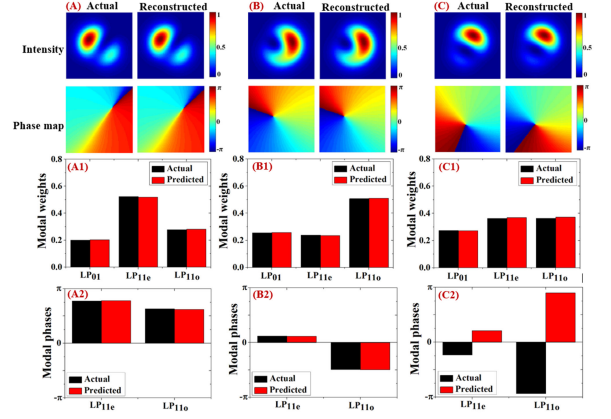


Fig. 4. Three typical testing results of the 3-mode case. The beam intensity and phase map are reconstructed based on the predicted mode coefficient, and the actual ones are also shown for comparison.

Fig. 4 shows three typical testing results of 3-mode case. The intensity or phase map reconstructed by the MD results are illustrated to be compared with the actual ones. It can be found that the reconstructed intensity of patterns (A)-(C) are of great similarity to the input actual ones, indicating high accuracy. As for the reconstructed phase map, (A) and (B) agree well with their actual one while (C) is opposite to the actual one, which results from the conjugated cases mentioned in Section II and it cannot be avoided since only one intensity image is utilized for decomposition. The far-field beam intensity can be observed to distinguish these conjugated pairs. The correlation between the actual and reconstructed pattern A, B, or C is 0.9995, 0.9998 and 0.9999 respectively.

We also compare the predicted and actual mode coefficients of these three typical samples in Fig. 4. The predicted modal weights of every eigenmode are very close to the actual ones for all the three samples, so as the modal phases of pattern A and B. In addition, the absolute value of predicted phases of pattern C is very close to the actual one, although the sign of them is opposite.

1000 simulated varying beam intensity profiles are processed by the trained CNN one by one and the evolution of the correlation between the measured and the reconstructed intensity pattern is shown in Fig. 5. Besides the correlation, six typical reconstructed and actual intensity images including the best and worst cases among the 1000 testing samples are illustrated in the insets of the figure. The corresponding frame number of these examples appears in the lower right corner of the measured intensity. The residual intensity patterns which can be calculated by $\Delta I = |I_r - I_m|$ and the corresponding correlation value are also provided. The averaged correlation of the investigated 1000 testing samples is 0.9987. Besides, it is found the total time to perform complete MD for these 1000 images is about 8 s, which means the decomposing rate for both modal weights and modal phases reaches about 125 Hz without considering the cost of capturing the beam intensity profiles. If we use the trained CNN to acquire only the modal weights, the decomposing rate can reach as high as 200 Hz.

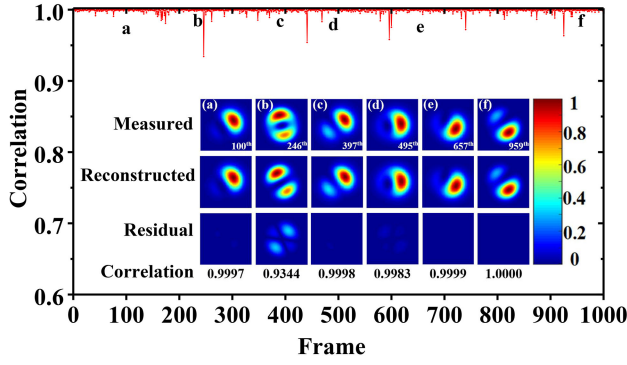


Fig. 5. The simulated real-time MD results for 3-mode case. The insets depict measured, reconstructed and the residual intensity of the frame denoted with letters. The best case is group (f) while the worst case is group (b).

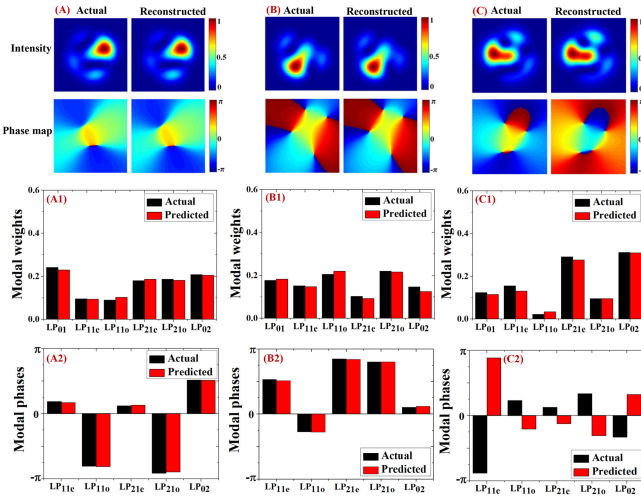


Fig. 6. Three typical testing results of the 6-mode case. The beam intensity and phase map are reconstructed based on the predicted mode coefficient, and the actual ones are also shown for comparison.

The similar analysis is also performed in 6-mode case. The typical testing results are shown in Fig. 6 while the evolution of correlation is shown in Fig. 7. In 6-mode case, the averaged correlation between the reconstructed beam patterns and the simulated measured ones is 0.9842. The total time to perform complete MD for 1000 sample images is about 28 s, which means the decomposing rate reaches about 36 Hz without considering the cost of capturing the beam intensity profiles. Similar to the 3-mode case, the decomposing rate for the modal weights is also about 200 Hz. The reason is that the time for CNN to generate an output vector for a beam pattern is about 5 ms whether in 3-mode case or in 6-mode case. However, to infer the modal phases from the output vector consumes much longer time for 6-mode case.

B. Experimental Results

With the experimental setup illustrated in Fig. 3, varying beam patterns in two different cases are recorded by the camera and sent to trained CNN frame by frame for decomposition

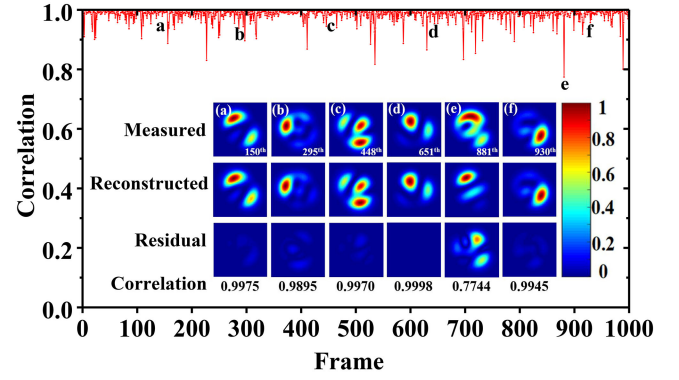


Fig. 7. The simulated real-time MD results for 6-mode case. The insets depict measured, reconstructed and the residual intensity of the frame denoted with letters. The best case is group (d) while the worst case is group (e).

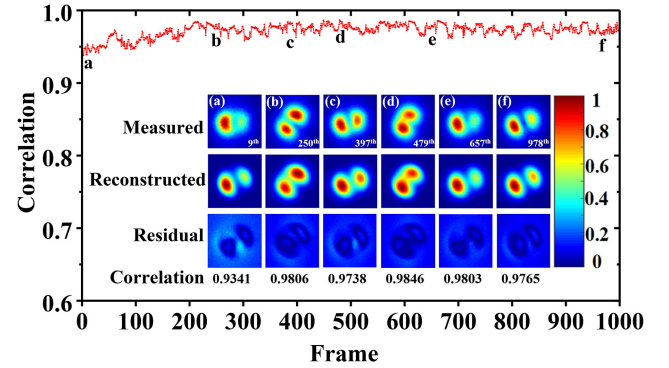


Fig. 8. The experimental real-time MD results for 3-mode case. The insets depict measured, reconstructed, and residual intensity of the frame denoted with letters. The best case is group (d) while the worst case is group (a).

respectively. The reconstructed patterns are found very similar to measured ones, showing the accuracy of our scheme.

Fig. 8 illustrates the evolution of the correlation between the measured and the reconstructed intensity pattern for 3-mode case. Six typical reconstructed and measure intensity images as well as the residual intensity images are also illustrated in the insets of the figure and their corresponding frames are denoted with letters. The correlation for these typical samples is texted below the corresponding patterns while the averaged correlation of the investigated 1000 testing samples is 0.9719. Noted the correlation is smaller than the simulated case, which is affected by the noise of the real beam patterns. Besides, the total time to perform whether complete MD or only modal weights determination for these 1000 images is about 33.4 s, which means the decomposing rate reaches about 29.9 Hz. To the best of our knowledge, this rate is the fastest one among the current numerical MD methods. Moreover, the real-time decomposing rate is restricted by the camera's maximum frame frequency, which is 30 Hz. We believe that a camera with a higher frame rate will achieve better real-time performance.

The evolution of the correlation for 6-mode case is shown in Fig. 9. The averaged correlation of 1000 frames are 0.8896. As for the cost time for complete MD, the total processing time is 41 s, which means the decomposing rate reaches about 24 Hz. If

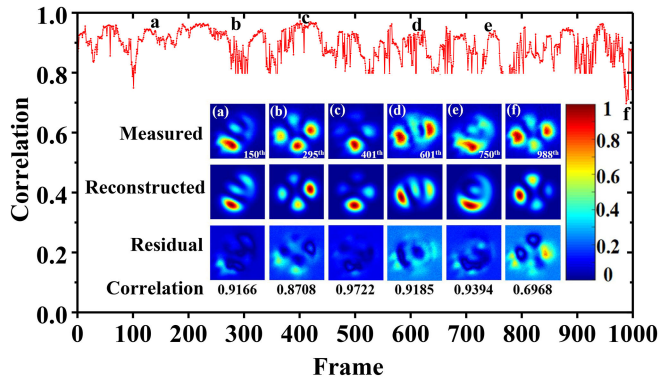


Fig. 9. The experimental real-time MD results for 6-mode case. The insets depict measured, reconstructed, and residual intensity of the frame denoted with letters. The best case is group (c) while the worst case is group (f).

the trained CNN is only used for determine the modal weights, the rate can reach as high as 29.9 Hz, restricted by the maximum of the frame rate of CCD.

From the results of two cases, we can find that the averaged correlation of the 6-mode case is much smaller than 3-mode case. The reason might be that 6-mode patterns are more complex than 3-mode patterns, so it will hinder CNN to accurately learn the relationship between the beam pattern and the modal coefficient. Besides, increase the training samples in 6-mode case might achieve better accuracy.

When it comes to higher-mode cases, we can infer that the correlation will further decrease and the MD error will be larger. To enhance the accuracy of higher-mode cases, we can try to adopt a deeper CNN, increase training samples in every training epoch, and optimize the learning rate, which will be a challenging work.

IV. CONCLUSION

We demonstrated the real-time ability of the deep learning-based MD method for MMFs on both simulation and experimental data. By using our method to acquire modal weights, the decomposing rate can reach as high as 200 Hz in the simulated case while 29.9 Hz in the experimental case. The real-time decomposing rate can be enhanced by utilizing a camera with a higher frame rate. Our method will also be beneficial for investigating unstable and varying beams of MMFs.

REFERENCES

- [1] D. Richardson, J. Fini, and L. E. Nelson, "Space-division multiplexing in optical fibres," *Nature Photon.*, vol. 7, no. 5, 2013, Art. no. 354.
- [2] P. Caramazza, O. Moran, R. Murray-Smith, and D. Faccio, "Transmission of natural scene images through a multimode fibre," *Nature Commun.*, vol. 10, no. 1, May 2019, Art. no. 2029.
- [3] L. G. Wright, D. N. Christodoulides, and F. W. Wise, "Controllable spatiotemporal nonlinear effects in multimode fibres," *Nature Photon.*, vol. 9, no. 5, pp. 306–310, 2015.
- [4] H. Defienne, M. Barbieri, I. A. Walmsley, B. J. Smith, and S. Gigan, "Two-photon quantum walk in a multimode fiber," *Sci. Adv.*, vol. 2, no. 1, 2016.
- [5] J. Carpenter, B. J. Eggleton, and J. Schröder, "Polarization-resolved cross-correlated (C2) imaging of a photonic bandgap fiber," *Opt. Express*, vol. 24, no. 24, pp. 27785–27790, Nov. 2016.

- [6] N. K. Fontaine *et al.*, "Characterization of space-division multiplexing systems using a swept-wavelength interferometer," in *Proc. Opt. Fiber Commun. Conf./Nat. Fiber Optic Engineers Conf. 2013*, Anaheim, CA, USA, 2013, Art. no. OW1K.2.
- [7] D. N. Schimpf and S. Ramachandran, "Polarization-resolved imaging of an ensemble of waveguide modes," *Opt. Lett.*, vol. 37, no. 15, pp. 3069–3071, Aug. 2012.
- [8] M. Mazur *et al.*, "Characterization of Long Multi-Mode Fiber Links using Digital Holography," in *Proc. Opt. Fiber Commun. Conf.*, 2019, Art. no. W4C.5.
- [9] R. Brüning, P. Gelszinnis, C. Schulze, D. Flamm, and M. Duparré, "Comparative analysis of numerical methods for the mode analysis of laser beams," *Appl. Opt.*, vol. 52, no. 32, pp. 7769–7777, 2013.
- [10] C. Schulze, D. Naidoo, D. Flamm, O. Schmidt, A. Forbes, and M. Duparré, "Wavefront reconstruction by modal decomposition," *Opt. Express*, vol. 20, no. 18, pp. 19714–19725, 2012.
- [11] Y. An, J. Li, L. Huang, J. Leng, L. Yang, and P. Zhou, "Deep learning enabled superfast and accurate M2 evaluation for fiber beams," *Opt. Express*, vol. 27, no. 13, pp. 18683–18694, Jun. 2019.
- [12] F. Stutzki, C. Jauregui, J. Limpert, and A. Tunnermann, "Real-time characterization of modal content in monolithic few-mode fibre lasers," *Electron. Lett.*, vol. 47, no. 4, pp. 274–275, 2011.
- [13] F. Stutzki *et al.*, "High-speed modal decomposition of mode instabilities in high-power fiber lasers," *Opt. Lett.*, vol. 36, no. 23, pp. 4572–4574, 2011.
- [14] I. Giles, A. Obeysekara, R. Chen, D. Giles, F. Poletti, and D. Richardson, "Fiber LPG mode converters and mode selection technique for multimode SDM," *IEEE Photon. Technol. Lett.*, vol. 24, no. 21, pp. 1922–1925, Nov. 2012.
- [15] J. W. Nicholson, A. D. Yablon, and S. Ghalimi, "Spatially and spectrally resolved imaging of modal content in large-mode-area fibers," *Opt. Express*, vol. 16, no. 10, pp. 7233–7243, 2008.
- [16] T. Kaiser, D. Flamm, S. Schroter, and M. Duparré, "Complete modal decomposition for optical fibers using CGH-based correlation filters," *Opt. Express*, vol. 17, no. 11, pp. 9347–9356, 2009.
- [17] Y. Z. Ma, Y. Sych, G. Onishchukov, U. Peschel, B. Schmauss, and G. Leuchs, "Fiber-modes and fiber-anisotropy characterization using low-coherence interferometry," *Appl. Phys. B*, vol. 96, pp. 345–353, 2009.
- [18] J. W. Nicholson, A. D. Yablon, J. M. Fini, and M. D. Mermelstein, "Measuring the modal content of large-mode-area fibers," *IEEE J. Sel. Topics Quantum Electron.*, vol. 15, no. 1, pp. 61–70, Jan. 2009.
- [19] D. Flamm, D. Naidoo, C. Schulze, A. Forbes, and M. Duparré, "Mode analysis with a spatial light modulator as a correlation filter," *Opt. Lett.*, vol. 37, no. 13, pp. 2478–2480, 2012.
- [20] D. Flamm, C. Schulze, D. Naidoo, A. Forbes, and M. Duparré, "Mode analysis using the correlation filter method," *Proc. SPIE*, vol. 8637, 2013, Art. no. 863717.
- [21] H. Lu, P. Zhou, X. Wang, and Z. Jiang, "Fast and accurate modal decomposition of multimode fiber based on stochastic parallel gradient descent algorithm," *Appl. Opt.*, vol. 52, no. 12, pp. 2905–2908, 2013.
- [22] J. Demas, "Sub-second mode measurement of fibers using C2 imaging," *Opt. Express*, vol. 22, no. 19, pp. 23043–23056, 2014.
- [23] L. Huang, S. Guo, J. Leng, H. Lü, and P. Zhou, "Real-time mode decomposition for few-mode fiber based on numerical method," *Opt. Express*, vol. 23, no. 4, pp. 4620–4629, 2015.
- [24] L. Huang, H. Lü, P. Zhou, J. Leng, and S. Guo, "Modal analysis of fiber laser beam by using stochastic parallel gradient descent algorithm," *IEEE Photon. Technol. Lett.*, vol. 27, no. 21, pp. 2280–2283, Nov. 2015.
- [25] L. Li, J. Leng, P. Zhou, and J. Chen, "Multimode fiber modal decomposition based on hybrid genetic global optimization algorithm," *Opt. Express*, vol. 25, no. 17, pp. 19680–19690, 2017.
- [26] M. Lyu, Z. Lin, G. Li, and G. Situ, "Fast modal decomposition for optical fibers using digital holography," *Scientific Rep.*, vol. 7, no. 1, Jul. 2017, Art. no. 6556.
- [27] Y. An, L. Huang, J. Li, J. Leng, L. Yang, and P. Zhou, "Learning to decompose the modes in few-mode fibers with deep convolutional neural network," *Opt. Express*, vol. 27, no. 7, pp. 10127–10137, Apr. 2019.
- [28] H. Gao, H. Hu, Y. Zhao, and J. Li, "A real-time fiber mode demodulation method enhanced by convolution neural network," *Opt. Fiber Technol.*, vol. 50, pp. 139–144, Jul. 2019.
- [29] W. Yan, X. Xu, and J. Wang, "Modal decomposition for few mode fibers using the fractional Fourier system," *Opt. Express*, vol. 27, no. 10, pp. 13871–13883, May 2019.
- [30] K. Choi and C. Jun, "High-precision modal decomposition of laser beams based on globally optimized SPGD algorithm," *IEEE Photon. J.*, vol. 11, no. 5, Oct. 2019, Art. no. 7103110.

- [31] D. Flamm *et al.*, "Fast M2 measurement for fiber beams based on modal analysis," *Appl. Opt.*, vol. 51, no. 7, pp. 987–993, 2012.
- [32] K. Xie *et al.*, "Adaptive phase correction of dynamic multimode beam based on modal decomposition," *Opt. Express*, vol. 27, no. 10, pp. 13793–13802, May 2019.
- [33] Y. An *et al.*, "Numerical mode decomposition for multimode fiber: From multi-variable optimization to deep learning," *Opt. Fiber Technol.*, vol. 52, 2019, Art. no. 101960.
- [34] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Represent.*, 2015, *arXiv:1409.1556*.

Yi An received the B.S. degree in optical information science and technology from the National University of Defense Technology, Changsha, China, in 2017, where he is currently working toward the M.S. degree with the College of Advanced Interdisciplinary Studies. His current research interest focuses on mode decomposition for fiber laser beam based on deep learning.

Liangjin Huang received the B.S. and doctoral degrees in optical engineering from the National University of Defense Technology, Changsha, China, in 2010 and 2016, respectively. He is currently an Assistant Research Fellow with the College of Advanced Interdisciplinary Studies, National University of Defense Technology. His current research interest focuses on mode decomposition on fiber laser beam and fiber design for high-power fiber lasers.

Jun Li received the Ph.D. degree from the University of Bonn, Bonn, Germany, in 2017. He is currently an Assistant Professor with the College of Advanced Interdisciplinary Studies, National University of Defense Technology, Changsha, China. His current directions of research interest include 3D content generation, 3D vision and computing imaging, etc.

Jinyong Leng received the B.S. and doctoral degrees in optical engineering from the National University of Defense Technology, Changsha, China, in 2004 and 2011, respectively. He is currently an Associate Research Fellow with the College of Advanced Interdisciplinary Studies, National University of Defense Technology. His current research interest focuses on high power fiber laser technology.

Lijia Yang received the B.S. and master's degree in condensed matter physics from Beijing Normal University, Beijing, China, in 1983 and 1986, respectively. She is currently a Professor with the College of Advanced Interdisciplinary Studies, National University of Defense Technology, Changsha, China. Her current research interest focuses on fiber laser and photonic crystal fiber.

Pu Zhou received the B.S. and doctoral degrees in optical engineering from the National University of Defense Technology, Changsha, China, in 2005 and 2009, respectively. He is currently a Research Fellow with the College of Advanced Interdisciplinary Studies, National University of Defense Technology. His current research interests include fiber laser/amplifier technology, coherent combining of fiber lasers/amplifiers, and adaptive optics.